1. Model Selection:

I selected Random Forest because our data has categories like Weather and yes/no descriptions like Accidents. It handles categories well and gives feature importance.

Features: Hour, Day, Car/Bike/Bus/Truck/Total, Weather , Accidents, Road Closed

Target: Traffic Situation

2. Evaluation as tester:

- Train 80% (4761 rows) / Test 20% (1191 rows)

- Used TimeSeriesSplit not random - because you can't train on future data that cannot be recorded.

- Accuracy: Correct / Total *100. For example 1050/1191 = 88.1% - Target ->85% PASS

- Confusion Matrix: If real = heavy but predicted = normal => CRITICAL FAIL

- Precision/Recall for heavy _80%

The usage of TimeSeriesSplit to train the bot not random split – since you can't train on future data that hasn't happened yet and therefore doesn't exist. Random split produces future info, TimeSeriesSplit keeps chronological order: training on past data and testing on the predicted future, like real logistics operation.

3. Test Results:

- TC2 Critical: 17:00 Moderate snow Accident =1 Road Closed = Yes -> Expected heavy + blocked --> PASS - safety works

- Overall 87.3% PASS

- Issue found: high class only 62% because there's only 380 samples – we need to use SMOTE. SMOTE creates synthetic new high samples by interpolating between existing high samples, so the model learns high traffic better without only just copying data.

4. Recommendation:

Hard rule: NEVER recommend if Road Closed=Yes even if model says normal.